

The Wide Signal

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COVER STORY

Does Creatine Increase DHT? What You Need to Know

Curious if creatine increases DHT? This popular supplement might boost DHT levels, a hormone linked to hair loss.

Greatist

Explore the connection between creatine, DHT, and male pattern baldness in our detailed guide.

Ah, creatine! It's a popular supplement promising muscle gains and enhanced performance. But as you're hitting the gym harder and seeing results, there are some potential side effects to be aware of. Here's the lowdown.

Creatine monohydrate is a naturally occurring substance found in muscle cells. It helps to produce adenosine triphosphate (ATP) — your muscles' go-to fuel when working hard.

Although you can find creatine in foods like red meat and seafood, many gym enthusiasts and athletes boost their stores with supplements. Why? Because more creatine equals more ATP, which means more energy for high-intensity activities. More fuel means more mileage!

Now, here's where things get hairy. You might dial up your body's DHT (dihydrotestosterone) production when you uplevel creatine. DHT is a derivative of testosterone and plays a big-league role in developing male characteristics. But here's the catch: while DHT might beef up your alpha vibe, it's also notorious for thinning out hairlines in peeps predisposed to male pattern baldness.

So, do creatine supplements cause baldness? Maybe! The buzz mainly stems from a small 2009 study where college-aged rugby players loaded up on creatine and saw their DHT levels increase significantly.

Because DHT can shrink hair follicles and shorten the hair growth cycle, it may seem logical to blame creatine for contributing to hair thinning. However, it's important to remember that these findings are a one-hit wonder — they haven't been seen again in other studies. Plus, let's

not forget that intense workouts alone can pump up these hormones, too.

As a result, the jury's still out, making the creatine-and-baldness saga a hot topic for gym gossip and scientists alike.

The same 2009 study found that after a week of taking high-dose creatinine at 25 grams per day, DHT levels rose by 56%. When the researchers dropped the dose to 5g daily for two more weeks, levels remained 40% higher than baseline. These findings contrasted with the control group participants who took a placebo.

Does this mean creatine could spell disaster for your do? Well, creatine boosts DHT, the usual suspect in male pattern baldness, also called androgenic alopecia. However, your disposition to hair loss and the effects of DHT is genetic, meaning that it's inherited.

Some folks are more prone to DHT's hair-ruffling effects because of how their androgen receptor gene is wired. Hormones like testosterone and DHT bind to androgen receptors, typically triggering normal hair growth. However, variations in the AR gene make the androgens extra effective, making you more likely to experience hair loss.

So, while there isn't much evidence to say that creatine supplements cause hair loss, the connection to increased DHT levels is worth considering if your family reunions look like a convention for bald eagles.

Beyond the potential for a breezy scalp, creatine is actually solid in the safety department, with long-term studies giving it a thumbs up. However, although creatinine doesn't

pose a risk to healthy individuals, it might not be your best move if you have kidney, high blood pressure , or liver disease.

Another thing to keep in mind is that creatinine loves to hang onto water , so expect some extra bloat and weigh-in surprises as your body adjusts.

Thinking of starting a creatine kick? Chatting with your doctor first is always wise, especially if you have other health conditions or take medications. They can advise you and rule out any potential interactions.

If you're taking creatinine and notice changes to your hair, a check-up is a good idea. Shedding a few strands is normal , but it's a red flag if you start losing clumps of hair . Rapid hair loss can indicate underlying health issues that go beyond your hairstyle.

Furthermore, consult a doctor if hair loss is weighing on your mind. They can offer solutions or treatments to manage your hair loss and stress .

Creatine is celebrated for its muscle-building prowess, but rumors of a link with hair loss exist. The good news is there's little conclusive evidence directly connecting creatine supplementation to permanent hair loss.

However, creatine loading may boost DHT levels, a hormone linked to male pattern baldness, especially in people genetically predisposed to hair thinning .

So, if you're concerned about hair loss, it might be wise to talk with a doctor before adding creatine supplements to your muscle-building routine.

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FEATURES

Martin Marshall: We need long-term, realistic, and tangible solutions to solve the crisis in general practice

julietwalker · The BMJ Blog

This week's announcement gave us a short-term sticking plaster that will do little to improve the morale of our beleaguered profession or the care we can deliver to our patients, writes Martin Marshall

The call for more face to face GP appointments have become part of the national conversation of late, so we expected GP access to be a focal point of this week's government announcement on NHS funding—but we didn't expect it to be the sole focus.

At the Royal College of General Practitioner's annual conference in Liverpool this week, the frustration and disappointment among our members has been palpable.

While £250m for general practice is a significant sum of money, it's the tip of the iceberg when you consider that this needs to be shared across over 7,000 GP practices in England. It may help get us through the winter—assuming that locums are available in areas of need—but achieve little beyond that.

Sharing some of the responsibilities for prescribing and writing fit notes is a welcome development, but falls a long way short of the system-wide programme needed to cut the box ticking and paperwork that fills up GP's days and takes us away from frontline patient care. Reducing the burden of bureaucratic QOF on practices would have been a more positive step forward.

Earlier this summer the RCGP published our own “recovery plan” for halting the crisis and protecting the care of our patients and the wider NHS by investing in the hardworking GPs and their teams who provide that care. Yet our realistic and tangible solutions have been largely ignored.

Every patient should have good access to a GP, but the government's obsession with access is a major distraction from the very real workforce and workload pressures that are preventing GPs from giving their patients the care they need and deserve.

It also fuels the rhetoric that GPs are “refusing” to see patients face to face which, as well as being inaccurate and unfair, undermines the bond of trust that GPs have with their patients.

GPs go into general practice to care for patients, and we share their frustrations when they experience long waits for GP appointments, but the claim that face to face consulting isn't happening is wrong. Almost six in 10 consultations are currently face to face—that's nearly 14 million in a month. The RCGP has always been very clear that a blend of remote and face to face

consultations are necessary, and that post-pandemic this should be a shared decision between GP and patient. We know some patients prefer to see their GP face to face—but good care can, and is, being delivered remotely and some patients prefer it.

Regardless of how our patients access GP services, they deserve safe care and there are limits beyond which we can no longer guarantee this, especially when an individual GP is now responsible for an average of 2000 patients—this is 5% higher than just six years ago.

Yet this week has also prompted the resurgence of allegations that long waiting times for a GP consultation are caused by GPs working “part time.” But if you actually look at the hours worked by a “part-time” GP working three days a week, they are longer on average than what would be considered full time by most people—around 40 hours. A quarter of GPs are working 50 hours a week or more. To put that into context, a pilot is restricted to flying 32 hours over seven days because doing more would be considered unsafe.

GPs have worked to their limits over the last 18 months and ensured the delivery of essential care and services to patients when other parts of the NHS shut down.

Despite this, hardworking GPs continue to be vilified and scapegoated from all directions. The plans to increase more scrutiny of hard-pressed practices and the introduction of an arbitrary text message service to rate the performance of particular GP practice will further demoralise a profession that is already on its knees. Ironically, this could lead to more hardworking doctors leaving the profession before their time.

The longstanding problems in general practice—resulting from more than a decade of under investment by successive governments in the family doctor service—have only been exacerbated by the events of the past 18 months and the supply of GPs is falling far short of patient need and demand, which is rising both in volume and complexity.

While previous experience has taught us not to expect a miracle cure, this week's so-called rescue package is yet another missed opportunity.

We need long-term, realistic, and tangible solutions to solve the crisis in general practice. This week's announcement gave us a headline-grabbing, short-term sticking plaster that will do little to improve the morale of our beleaguered profession or the care we can deliver to our patients.

Martin Marshall, Chair of the Royal College of GPs, a GP in East London, and Professor of Healthcare Improvement at UCL.

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Charlotte Blease: We need to talk about “closed notes”

julietwalker · The BMJ Blog

Lack of easy access to personal health records is associated with adverse consequences for patients, writes Charlotte Blease

“The cancer might be stage 1b. Then again it could be 3b.”

“Let’s check Google again.”

This was the patter of daily, and nightly, conversations between my partner, H, and myself over a two-week period. Diagnosed with stomach cancer, H could recall scarcely anything the oncologist said. To fill in the knowledge gaps, we did what every doctor advises against—we turned to Dr Google. H also requested copies of his medical records—several times. As we waited, worried, and felt confused, the internet offered no concrete answers, but it did help us claw back some semblance of control. At least now we had information on the treatments and prognoses for stomach cancer—even if we didn’t yet know the details of H’s case.

Not everyone with a cancer diagnosis experiences the frustration of not being able to look at their medical notes and think through the implications of the information documented there. In the USA and Nordic countries, patients are offered real-time, online access to their clinical records—including lab and test results and their doctors’ notes (“open notes”). Recently, much attention has focused on the implications for patients and clinicians of this practice innovation. But another conversation has been overlooked. Denying patients’ online access to their medical information carries consequences too. We need to talk more about “closed notes.”

Legally, patients in many countries have the right to request copies of their medical records. However, as my partner and I experienced, the process is cumbersome, time-consuming, and socially awkward. At a time when he should have been thinking only of himself, H was worried that his doctor would think negatively of him for daring to seek more information about his condition. Emails to his busy doctor went unanswered, and this added to his nervousness. Closed notes unquestionably heightened his anxiety, and intensified confusion about the future. H recognized his doctor was overstretched, but felt worthless and frustrated when nobody had time to reply. By obtaining information about our sensitive internet searches about stomach cancer, Google seemed to be the only winner in this scenario.

H’s experiences are not unique. Patients misremember around half of what their doctor tells them, and recall is worse when patients are in shock. Studies show that “scanxiety”—the distress felt by patients waiting for test and scan results—is real, and dissipates when patients get their results. Among patients who lack ready access to their results and clinical notes on how to interpret them, the wait is protracted and stressful. In the current system,

however, doctors don’t tend to focus on the adverse effects on patients of “closed notes.” Perhaps this is because it is the current convention in most parts of the world. And patients seldom complain, as most are resigned to a lack of access to personal health information and test results because it’s long been “how things are done.”

Emails to his busy doctor went unanswered, and this added to his nervousness.

julietwalker

Reflecting on this customary practice invites important ethical questions. Closed notes are reliant on single-shot, miss-it-and-you-miss-out disclosure of information by doctors to their patients. While most are now copied into clinical letters sent to their primary care doctors, not all are and there is no onus on doctors to write notes which are easily understood by patients. How does this affect patient empowerment, wellbeing, or patient-doctor trust? What is the impact in terms of patient harm? And what do closed notes do for justice in healthcare delivery?

H was in a relatively privileged position. But despite being well-educated and fluent in English, he still struggled to remember everything his doctor said. Among patients from other demographic groups—minorities, people with low-incomes, older people, those suffering from multiple chronic conditions, or patients who don’t speak the same language as their doctor—poor communication and misunderstanding is even more likely. Patients from lower socio-economic groups, and those from racial or ethnic minorities, may experience anxieties related to perceived or anticipated prejudice potentially further compromising engagement and medical dialogue. Older patients may suffer from impaired memory making it harder to recall recommendations and next steps. People with multiple chronic illnesses may simply have too much information to remember. Patients whose first language differs from their doctor’s are also at higher risk of misunderstanding their diagnosis and treatment, resulting in treatment non-adherence and adverse reactions to medications.

Open notes are the solution to important problems that have been overlooked and downplayed by health professionals. Affording patients digital access to their own health information allows them to extend their engagement with the information provided at the clinic visit, and supplement the communication pitfalls of time-constrained face-to-face or virtual appointments.

In multiple studies, the majority of patients with experience of open notes report that reading their notes is very important for feeling in control of their care, remembering their care plan, and understanding why medications are prescribed. Patients report trusting their provider more, greater goal alignment, and teamwork. These benefits are reported most often among patients who are older, non-white, those with fewer years of formal education, or do not

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Extra Skin? 9 Exercises for A Sagging Neck

Greatist

Saggy neck? Tighten things up with targeted moves for “turkey neck.”

We’re not in love with the term “turkey neck” — it’s just extra skin.

If you don’t want to learn to love the extra chin you’re in (though we wish you would!), you can perform some exercises that may tighten your neck muscles.

AleksandarNakic/Getty Images

Though some swear these moves work, there’s no research backing them up, and they can’t zap away extra flowy skin.

Instead, they’re more likely to help strengthen the muscles underneath loose neck skin. Unfortunately, skin removal is usually accomplished with more intense methods, like surgery .

However, these popular neck exercises are simple enough for a test drive and may help tone your neck muscles naturally.

1. Pucker up

Sit comfortably in a chair, back straight.

Tilt your head all the way back, chin fully tipped to the sky.

Pucker up nice and big, like you’re giving a selfie duck face or are fully making out with your invisible crush. If it helps, miming “mwah” with your mouth or saying that sound out loud in an exaggerated facial movement will get the job done.

Repeat up to 20 times if it’s not too uncomfortable.

2. Get chewy

Get in a comfortable sitting position with a straight back again.

Gently let your head fall all the way back and aim your chin up toward the ceiling.

With your mouth fully closed, move it around like you do when chewing a particularly gummy meal, just remember to keep both lips closed.

Perform this fake chewing sesh about 20 times if you can.

3. Sit-ups for your neck

Get in bed and lay flat, with your head hovering over its edge.

Bring your head up toward the ceiling with slow and intentional movement, engaging only your neck muscles.

Perform about 5 of these neck-ups if you can.

Remember to be gentle with this move. Though this neck lift exercise is simple, you don’t want to pull a neck muscle.

4. Frowning

Curl your mouth edges down into a big frown (if you’re unafraid of frown lines).

The more animated the movement, the better — you should feel some resistance.

Repeat a few times.

5. Facepalm

Put your palm on your forehead, like you’re Ben Affleck-level stressed out .

Try to drive your head into your hand, but don’t let your head move forward at all — like you’re holding yourself back from a fight.

Pause in this resistance for nearly 10 seconds.

Using both hands, hold the back of your head.

Using your neck, push backward, pausing for 10 seconds.

6. Chin to chest (neck flexion)

Stand straight, arms by your sides, core engaged .

Bring your chin to your chest while bringing your shoulders back and down behind you.

Pause here for 30 seconds before performing a handful of times.

7. Fish face

Take a nice deep breath.

After the exhale, suck in your cheeks as much as you can so your lips are absolutely puckered out.

Keep this look for 5 seconds or longer before performing it up to 10 times.

8. Baby bird facial yoga

Push your tongue tip through the top of your mouth as much as you can while smiling .

Keeping that tongue smile secure, look all the way up to the ceiling 10 times.

9. Closed mouth smile

Grin as wide as you can with a shut mouth.

Keep your lips as close to your teeth as much as you can so

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Eating More Ultra-Processed Foods Could Damage Your Bones. Here's Why

Healthline

A new study suggests that people who eat more ultra-processed foods have poorer bone health. Sibila and Pavel/Stocksy

Eating higher amounts of ultra-processed foods is associated with lower bone mineral density, even in younger adults under 65.

Researchers also found that high ultra-processed food intake increases the risk of hip fractures.

Experts say the lack of nutrients needed for bone health in ultra-processed foods could contribute to the association .

Eating more ultra-processed foods is associated with poorer bone health.

Research published in The British Journal of Nutrition found that those who eat larger amounts of ultra-processed foods (UPFs) had a higher risk of hip fractures and lower bone mineral density.

The findings were pronounced even among younger adults under 65 and in those who were underweight.

“Our study cohort was followed for over 12 years, and we found that high intakes of ultra-processed foods were linked to a reduction in bone mineral density at several sites including key areas of the upper femur and the lumbar spine region,” Lu Qi , MD, PhD, co-author of the study and HCA Regents Distinguished Chair and professor at the Celia Scott Weatherhead School of Public Health and Tropical Medicine at Tulane University said in a press statement .

“While recent studies have shown that ultra-processed food consumption can affect bone health, this is the first time this relationship has been examined directly in humans,” Qi continued.

Ultra-processed foods weaken bones, raise hip fracture risk

Data from the CDC suggests that ultra-processed foods account for roughly 55% of total calories consumed by both children and adults.

To explore the impact of ultra-processed foods on bone health , the researchers analyzed data from more than 160,000 participants in the UK Biobank database.

On average, people ate roughly 8 servings of ultra-processed foods per day.

The researchers found that for every 3.7 additional servings of ultra-processed foods consumed each day, the risk of hip fracture increased by 10.5%.

The study authors note that while the serving sizes of ultra-processed foods can vary by food type, 3.7 additional serves equate to a frozen dinner entree, a soda , and a cookie.

Grace Derocha , a registered dietitian nutritionist and national spokesperson for the Academy of Nutrition and Dietetics, said it's a finding worth paying attention to. Derocha wasn't involved in the study.

“A 10.5% increase in hip fracture risk is meaningful, especially given how serious hip fractures can be for long-term mobility and independence, particularly in older adults,” she told Healthline. “That said, it is important to interpret this in context. This is an observational finding, meaning it shows an association rather than direct causation.”

“Still, it reinforces a pattern we see across nutrition science: diets higher in ultra-processed foods tend to be linked with poorer health outcomes overall. From a public health standpoint, it's a signal worth paying attention to — not necessarily a reason for alarm, but certainly a reason to emphasize improving overall diet quality,” Derocha said.

Dana Hunnes , a senior clinical dietitian at UCLA Health, said there could be several mechanisms underlying the association between ultra-processed foods and poorer bone health. Hunnes wasn't involved in the study.

“Inflammation could potentially be a driver of lower bone density; lower intake of calcium, phosphorus , vitamin D, and other minerals important to bone health may be consumed less often when intake of ultra-processed foods [is] consumed more often. Without sufficient amounts of these healthy nutrients, bone density may weaken,” she told Healthline.

“Perhaps people are also less active, and less activity also can reduce bone density, as you're not having the same compression or forces on the muscles/bones as you are when you're more active. All of these factors may also be combining to affect this change,” she explained.

Younger adults, those with lower BMI also at risk

The association between ultra-processed foods and lower bone density was most pronounced among those under age 65 and in those who were underweight with a body mass index (BMI) of less than 18.5.

The authors note that these associations could be more pronounced in younger people due to stronger digestive function, which could lead to better absorption of the unhealthy ingredients in ultra-processed foods.

Those with a low BMI are also already at risk for bone health issues, and this may make the impact of ultra-processed foods on bone density worse.

How diet impacts bone health

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Judge Blocks RFK Jr.'s Child Vaccine Policies, Says They Disregard Science

Healthline

A federal judge has overturned HHS Health Secretary RFK Jr.'s changes to the childhood vaccination schedule. Matthias Bein/picture alliance via Getty Images

A federal judge has struck down childhood vaccination recommendations implemented earlier this year by the U. S. Department of Health and Human Services (HHS).

The judge also blocked HHS Health Secretary Robert F. Kennedy Jr.'s appointments to the Advisory Committee on Immunization Practices.

Medical professionals praised the judge's ruling, saying vaccinations are vital in preventing diseases in children.

A federal judge has overturned new childhood vaccination policies implemented earlier this year by the U. S. Department of Health and Human Services (HHS).

U. S. District Judge Brian Murphy ruled that federal health officials under the leadership of HHS Health Secretary Robert F. Kennedy Jr. had acted unlawfully when they issued new childhood vaccination recommendations in January.

Those guidelines reduce the number of childhood vaccinations, indicating vaccination against 11 diseases instead of the 16 diseases recommended under prior guidelines. The HHS policies also downgraded childhood immunization recommendations for other diseases, including rotavirus, influenza, and hepatitis A.

In his ruling, Murphy said that previous childhood vaccination recommendations had been made through "a method scientific in nature and codified into law through procedural requirements."

The judge said that HHS officials had "disregarded those methods and thereby undermined the integrity of its actions."

Murphy also ruled that Health Secretary Kennedy's appointments to the Advisory Committee on Immunization Practices (ACIP) were not lawfully constituted, and blocked Kennedy's 13 appointees to the panel.

The plaintiffs in the case argued that the committee had become dominated by people aligned with Kennedy's anti-vaccine views and was constituted in violation of the Federal Advisory Committee Act's mandates that it be fairly balanced and free of inappropriate influence.

After the ruling was issued on March 16, ACIP meetings scheduled for this week were postponed. The panel was scheduled to discuss potential changes to recommendations regarding COVID-19 vaccines.

In addition, the judge had put a hold on votes taken by ACIP members since June. That includes a December decision by the panel to roll back recommendations that newborns receive a first dose of the hepatitis B vaccine within 24 hours of birth.

The Trump administration is expected to appeal Murphy's ruling.

Medical professionals praise vaccination ruling

The lawsuit challenging the HHS childhood vaccination recommendations was brought by the American Academy of Pediatrics (AAP) and other major medical groups.

"Today's ruling is a historic and welcome outcome for children, communities, and pediatricians everywhere," said Andrew Racine, MD, AAP president, in a statement.

This decision effectively means that a science-based process for developing immunization recommendations is not to be trifled with and represents a critical step to restoring scientific decision-making to federal vaccine policy that has kept children healthy for years," Racine continued.

Medical professionals said the judge's ruling was an important action and the correct decision.

"The judge's ruling brings light and focus to the reality that the changes to the vaccine schedule were made by individuals who are not experts in vaccinations, science, or public health and the changes were not based upon any new data, evidence, or scientific basis," said Graham Tse, MD, a pediatrician and chief medical officer of MemorialCare Miller Children's & Women's Hospital in Long Beach, CA.

"The judge's ruling is everything that pediatricians and family practice physicians and the AAP have been waiting to hear," said Danelle Fisher, MD, a pediatrician at Providence Saint John's Health Center in Santa Monica, CA.

"There was absolutely no reason to downgrade recommendations on vaccines. The ones who will suffer are the children. Someone needed to stand up to this administration and help our kids," she told Healthline.

Tse said the judge's ruling, if upheld on appeal, will send a clear message to parents and other community members.

"I would hope this brings back a single recommended vaccination schedule, supported by all states and the federal government, for the United States," he told Healthline. "When there are vaccine schedule disagreements and variability across agencies, groups, and states, it leads to confusion, fear, anxiety, and ultimately decreased vaccine acceptance."

Why childhood vaccines matter

The AAP recommends that children receive vaccinations for 18 different diseases from birth through age 18.

These inoculations include protection against:

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Single use plastic in healthcare must not become the new normal

julietwalker · The BMJ Blog

We must not let our reliance on single-use plastic in healthcare become the “new normal” or set-back the strides taken prior to the covid-19 pandemic to address the primary existential crisis concerning our environment.

The covid-19 pandemic has highlighted the huge material footprint required to support frontline healthcare in the NHS. The NHS contributes 5.4% to the UK's climate footprint. [1] Pre-covid, the NHS had made great strides in tackling waste and contributing to its goal of a net-zero future. In 2019, the NHS had achieved an estimated 62% reduction in its 1990 emission levels. [2]

However, six months after the initial outbreak of covid-19, the UK government had procured 32 billion items of personal protective equipment (PPE) worth £12.5 billion. [3] Between February 2020 and February 2021, the NHS used more than 1 billion surgical masks. [4] This would generate nearly 3 million kilograms of waste according to a recently published study, which estimated that 123,000 tonnes of unrecyclable plastic waste would be generated in a year if every person in the UK used one disposable surgical mask per day. [5]

To support the long term plan of achieving net-zero and other sustainability commitments, more systemic approaches to designing out waste and greenhouse gas emissions are needed within the NHS. Current practices of incineration, landfill, and low-value recycling need to be replaced by a “circular” mind-set, where everything is designed, procured, and used for as long as possible, at their highest value, only then recycled back into new products or for alternative high-quality uses. [6] Such approaches are available even during a pandemic.

For example, a Sterimelt machine, which aims to recycle and repurpose wasted surgical masks and other plastic, was installed during the pandemic at the Royal Cornwall Hospitals NHS Trust. [7] A recent study found that the carbon impact of anaesthetic gases can be as small as intravenous anaesthetics, by shifting from desflurane to sevoflurane and employing effective vapour capture technologies. [8] A trial of world's first bespoke net-zero clinical laundry study for Personal Protective Equipment (PPE) piloted across a range of NHS sites for gowns, aprons, coveralls and surgical masks found that the service has the potential to save the NHS an estimated £9m per annum, displacing 5% of the disposable market share in 2022. [9] At a product level, the re-manufacture of electrophysiology catheters was found to reduce its carbon emissions by a minimum of 50% [10]; while a “circular” mindset led to the development of a technology that can transform materials of clinical wastes into feedstock chemicals. [11] At an organisational level, monitoring and increasing the use of building space, shifting to performance-based business models for expensive equipment and lighting, and altering procurement rules to favour remanufactured products are all actions that are needed to

create large scale system changes to break out of the current linear model and ways of thinking.

However, the human element is crucial during this transition. Continuous monitoring and widespread education and awareness training are needed to ensure the correct choices are made and implemented on a daily basis. A recent study highlights that the long list of factors that have contributed to the embedding of single-use practices and norms in US healthcare including: just in time purchasing systems to minimise storage; planned obsolescence from manufacturers and suppliers; mistaken belief that re-use impacts risk and patient safety; concerns about liability and cost; confusing classificatory and asymmetric regulatory frameworks; and split roles and incentives for taking action. [12]

Currently, most medical sites do not have a post-consumption treatment system for composting plastics. The popularity of single-use compostable plastics in a pandemic is understandable, but concerning, as single-use plastic could become a regrettable and expensive option. [13] While staff perceptions of what products can and cannot be recycled in the local area is an on-going issue within the NHS irrespective of the covid-19 pandemic, re-use approaches would remove much of the complexity and confusion about legacy recycling models.

Some may consider that the plastic crisis caused by the need to use large volumes of PPE during the pandemic is an environmental “price worth paying” during an unprecedented public health emergency; however, the reality of the climate and environment emergency means a step-change is required across the NHS irrespective of the global pandemic. While the process will be difficult, evidence from other resource and asset-heavy sectors demonstrates the positive business cases and low costs of a circular economy approach towards net-zero. [14] It requires leadership from the NHS to hold the corporate and suppliers accountable, and the ambition to create large scale improvement.

Peter Hopkinson is the co-director of the Exeter Centre for Circular Economy.

Richard Smith is the inaugural deputy pro-vice Chancellor for the University of Exeter Medical School.

Lora Fleming is based at the European Centre for Environment and Human Health (University of Exeter Medical School).

Karyn Morrissey is a Professor at the Department of Technology, Management and Economics Sustainability, Technical University of Denmark.

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Competing interests : none declared.

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Healthcare in our hands: Putting babies, children, and young people at the centre of their care

kellybrendel · The BMJ Blog

Accessing healthcare services can be daunting at any age, but even more so as a baby, child, or young person. Can you imagine being told life changing information and not having any idea of what's going on? It's a difficult and help-less position to be in, and we felt overwhelmed, anxious, and vulnerable when we experienced this as children and young people. When all your decisions are made for you, albeit with the best of intentions, and you don't have much involvement apart from receiving the treatment, it's easy to feel fearful and powerless. Yet everyone is entitled to a say in the healthcare services that they use and a child or young person is no different.

The United Nations Convention on the Rights of the Child states every "child who is capable of forming his or her own views has the right to express those views freely in all matters affecting the child." The National Institute for Health and Care Excellence (NICE) recently published guidance on Babies, Children, and Young People's Experience of Healthcare, and this principle has been embedded throughout the guideline. By actively involving the views of children, young people, and the parents and carers of babies and young children, this guidance truly represents those who use the service and will mean it can make a real difference.

The recognition that children and young people have a right to share their perspectives is not only woven throughout the recommendations, but was also remembered all the way through the guideline development. From scope development to publication, young people with personal experiences of accessing healthcare have been involved in this guideline.

Having personal experience of accessing healthcare gives you a unique viewpoint that is not always heard, especially if you are from under-represented groups, including those who are under 18, disabled, from an ethnic minority background, or who don't have English as a first language. It was vital for these voices and experiences to be heard, listened to, and then put into this guidance to ensure it can improve care for as many people as possible. The guideline committee included four young people with varying levels of experience accessing healthcare (including two of us, Aishah and Emma). Having this wealth of experience to draw on meant that we were able to bring our expert opinion and real life testimonies to the guideline committee meetings, rather than just relying on examples from research or journal articles.

The guideline has formed recommendations in a broad range of areas, but we want to highlight two in particular. First, communication. We know from our experiences that often when our parents and carers are present, healthcare professionals will direct all communication towards them, even though it's about us. This can be disempowering and risks us disengaging with our healthcare (for example, by

being reluctant to access services or adhere to treatment). It's essential that we feel included in conversations and that our preference of communication is understood.

It is so important to engage with and provide information to a child or young person in a form of communication that they feel comfortable with, whether that be through Makaton, pitching the language used at the right level, or using additional visual aids for support. It can help them feel more at ease, builds trust, and opens up more opportunities for them to be involved in their own healthcare journey.

Shared decision making is an important part of any individual's healthcare journey—especially a child or young person's. At this age we can have little involvement and voice in our healthcare, but shared decision making helps to position us in the centre of our care. The NICE guideline sheds light on how children and young people can be supported to make decisions, even if they're as simple as choosing a plaster cast colour or food options on a menu. As a child and young person, we've both had similar options chosen for us, but when we knew what we wanted, why weren't we asked? Being involved in simpler decisions that may not have a direct impact on our healthcare outcome can seem unimportant, however it's crucial, because it affects our overall perception of healthcare and builds our confidence to engage in bigger healthcare decisions. It is also well known that the transition from paediatric to adult services can be a really difficult time for patients and it's not hard to see why: how can we be responsible for our own care if we haven't had the opportunity to learn how to do it?

As part of the evidence for our recommendations, NICE commissioned the National Children's Bureau to run focus groups with 200 children aged 4-15 years old. To conclude, we'd like to include their voices about how they felt being involved: "Young person's voices should be listened to, valued, and considered to ensure that the health service can be accessible to all"; "The best thing is giving our opinions and being part of something important"; "I like you listening to my views, it's awesome!"; "I really like that it feels that you all care about my opinion."

Through this guideline, we saw that it was possible to fully engage with children and young people. Now it's time to put that into practice across healthcare.

Aishah Farooq , committee member of the NICE guideline on Babies Children and Young People's Experience of Healthcare; young governor for University Hospitals Bristol and Weston; NHS England Youth Forum alumni; NHSE public patient voice partner for the asthma workstream. Twitter @ItsAishahF

Emma Beeden , committee member of the NICE guideline on Babies, Children, and Young People's Experience of Healthcare; NHS Youth Forum alumni; member of the British Medical Association Patient Liaison Group. Twitter @emmabeeden

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Text Neck Exercises: Simple Moves to Ease Digital Strain

Greatist

Some gentle stretches and changes to your device habits can help alleviate neck pain.

Whether you're caught up in doomscrolling or just checking your work emails, chances are you're spending a lot of time on your phone. But all that screen time can lead to a pesky condition called "text neck." Hunching over your device can leave you with some severe aches and pains.

So, hold your phone up to eye level and check out these easy tips to get rid of text neck fast!

AleksandarNakic/Getty Images

Also known as "tech neck," text neck is a term coined by a chiropractor to describe the neck pain you might feel after spending too much time looking down at your phone.

Most of us check our phones with our heads tilted forward and down. When you do this for too long, the muscles in your neck get strained. The muscles in the back are over-stretched, while the ones in the front are compressed.

If you have text neck, here's what you might be dealing with:

Stiffness and difficulty moving your neck

Dull aching or sharp, stabbing pain

Pain that radiates into your shoulders and arms

Weak shoulder muscles

Tension headaches from tight neck muscles

But don't get bent out of shape about overusing your favorite device. There's no solid proof that your phone habits are to blame for your neck woes. Several studies found no connection between using your device and having neck or shoulder pain .

You could get a sore neck from overusing your muscles for many other reasons, like genetic factors, weight issues, or bad posture. "Text neck" might be a term concocted by a chiropractor to get your business. But, hey, we're not judging.

To help ease text neck pain, try these simple exercises.

Neck circles

While sitting, slowly rotate your head 360 degrees.

Let your jaw slack as you roll backward, and be very gentle.

Rotate your head clockwise five times, then counterclockwise five times.

Again, go VERY slowly (like a full ten seconds for one rotation), and don't force anything!

Side neck stretch

Take your right hand and reach over your head to the top of your left ear.

Use the weight of your arm to guide your head toward your right shoulder very slowly. (Don't engage your neck muscles at all!)

Remember to breathe deeply, even if your armpits are ripe.

When you feel discomfort, pause and breathe again, letting your head stay there for three full breaths.

Gently remove your hand from your head and use it to push your head slowly back up to an upright position.

Repeat on the left side.

Shoulder rolls

Slowly raise your shoulders.

Without stopping, roll them back and down.

Do 10 to 15 reps.

Repeat the exercise, but roll your shoulders forward and down this time.

Bonus: Rolling your shoulders will help with your dance moves and text neck.

Chin tucks

Gently tuck your chin towards your chest, like you're making a super sexy double chin. Bonus: Smile like Kristen Wiig while doing this part.

Keep your head level — no tilting or bending!

Hold that position for 5 seconds (or the length of a deep breath if counting isn't your thing).

Relax your neck and repeat.

Aside from these cubicle-friendly exercises, here are some ideas to help keep text neck from being a regular pain in the y'know:

Hold your phone, tablet, or device a bit higher to reduce your head and neck stress.

Try yoga , pilates , or other activities that engage your core , lengthen your spine, and loosen up your neck .

Take breaks from your phone, laptop, or any other device that might make you hunch over.

Move around while you're taking your break.

Neck circles While sitting, slowly rotate your head 360 degrees.

Greatist

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The AI boom has plunged a small Pennsylvania town into chaos

Rebecca Egan McCarthy · Grist

class="has-drop-cap has-default-font-family">"I don't like to see anyone upset," said Nick Farris of Provident Real Estate Advisors. He was sitting in the front of a crowd of roughly 150 inside Valley View High School's auditorium in Archbald, a town of about 7,500, huddled between two mountain ranges in Pennsylvania's Lackawanna Valley. Farris was there to represent the developer for Project Scott, one of many data center campuses coming to town. "However," he said. "I think that this is the best data center site in this area of the country, by far." The audience had been fairly quiet, bundled in thick coats against the late January cold. But as Farris spoke about data centers as a boon for communities, they began to laugh, drawing a rebuke from town officials.

class="has-default-font-family">"What about the children?" someone shouted from the crowd. The children were watching from the walls; long banners of Valley View Performing Arts students hanging around the auditorium like championship pennants. Project Scott and four other data facilities will sit just a few thousand feet from the middle and high schools.

class="has-default-font-family">"Isn't there a missile plant next door?" Farris said, getting aggravated. He was referring to Lockheed Martin's 350,000-square-foot Missiles and Fire Control facility directly next to the high school, parts of which are highly contaminated.

class="has-default-font-family">"That sucks too!" another attendee yelled back. This was nothing to worry about, Farris tried to convince the audience. This would bring in tax revenue, he said. It was just an office park, albeit one with roughly 450 diesel backup generators.

class="has-default-font-family">"It's going to be away from everyone," Farris kept repeating, to rising jeers. He was wearing a knit turtleneck with a large American flag emblazoned across the chest. "It's not going to bother anyone."

class="has-default-font-family">The specifications say something different: Five developers are planning to build six data center campuses in Archbald, which will cover a full 14 percent of the town, evict a trailer park, and border many residential properties. One campus alone, as The Scranton Times-Tribune reporter Frank Lefneskey pointed out, is expected to use more power than the region's largest power plant is able to produce.

class="has-default-font-family">Pennsylvania has become an epicenter of the data center boom in the United States, with over 50 campuses in development. Eleven of them are slated for Lackawanna County alone. Archbald, with six campuses composed of 51 massive buildings, has the most of any municipality in Pennsylvania.

class="has-default-font-family">Despite the public outcry, it has been surprisingly difficult to learn what Archbald's

elected officials think of the massive industry moving in. None of the town's seven council members responded to my emails, so I stopped by the borough administration building in person a few weeks before Christmas, and was told to wait in the lobby while they held an informal closed meeting in council chambers. When the door opened, it was clear that anyone who actually had the power to make or break the data center plans had quietly filed out the back, leaving me with Archbald's unelected borough manager, Dan Markey, who essentially runs the town, but cannot vote for or against development.

class="has-default-font-family">What did he think about artificial general intelligence, or AGI — the idea that eventually these efforts will produce something like a "computer god" capable of solving climate change, ending hunger, revealing the full breadth of science, and performing any number of other miracles? "I believe in one God, and it's not a computer," Markey told me evenly. By now, hundreds of towns across the country have been caught up in the rush to build large language models that can parse unfathomable amounts of data to write copy, answer any query, develop new vaccines, and likely render a large number of jobs obsolete. That rush accelerated into an all-out arms race over the past year. What it means in practice is an enormous amount of data centers — enough to approximate a minor deity and enough, as OpenAI co-founder and former chief scientist Ilya Sutskever once speculated, to "cover the Earth." Markey told me he had ChatGPT on his phone, but didn't use it much.

class="has-default-font-family">"I don't think anyone in their right mind wants to see the world covered in data centers," Markey said. "[But], according to Pennsylvania law, we have to have a zone for everything. An adult bookstore, a strip club, a concrete and asphalt plant — anything that wants to come here. We have to have a zone for it. If it's not zoned, it's allowed to go anywhere."

class="has-default-font-family">In most states, towns have the right to exclude businesses that they find disagreeable — a wealthy suburb, for example, would likely reject a landfill or gas plant — but in Pennsylvania, towns must allocate some patch of land to these "undesirable industries." Some municipalities deal with this by forming what are called zoning collaboratives, which allow them to plan as a region for pollution. One town might get data centers, another gas plants, another a landfill. Markey approached the nearby boroughs of Blakely and Dickson City to discuss the possibility, but the data center development rush has outpaced him.

class="has-default-font-family">Whatever's attracting data centers to the area, it's forced Markey to answer for a rush of development, unprecedented since the town was settled in the 1840s in service of an industry that will bring vanishingly few jobs. Residents kept bringing up the threat of collapse — the network of empty mine shafts running underground, the town beneath the town, the structural instability that would accompany these massive buildings. The data centers would drive homeowners into town, they said

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Embracing Afghanistan's refugees into the UK's health and social care system

julietwalker · The BMJ Blog

The response to refugees arriving in the UK after the international emergency in Afghanistan requires health professionals to be equipped for the needs of this population. The integration of this community from a health and social care perspective is key in terms of ensuring equitable provision. The Equality and Diversity Act (2010) must frame this planning, from where services can monitor and embrace this community to ensure they are not being disproportionately impacted. [1] The UK has committed to accepting 20,000 refugees over five years in response to the political takeover by the Taliban in Afghanistan. Healthcare professionals will need to develop allyship and effective working practices including challenging discrimination, stereotypes, racism, and xenophobia.

There must be relevant training and resources within health and social care teams and departments to ensure that refugees' needs are met and communication is safe. The Cultural Atlas (2021) provides one of the most comprehensive educational tools in understanding the culture and guiding healthcare practice on etiquette and communication. [2] The lack of available acceptable housing is going to be one of the largest issues facing Afghan refugees. There needs to be a close working relationship between community health services, homeless teams, and housing services. Such relationships are key as there is a duty of care to anyone in the local area to be housed even if that is in temporary accommodation. Health professionals may be the first responder in ensuring that local signposting to organisations and advocacy agencies are appropriately done.

Another issue that will need to be addressed in a timely fashion is the lack of money and finances. Contrary to myths surrounding refugees and asylum seekers who enter the country, they will not have money to live the lifestyle alleged by many. An asylum decision takes at least 28 days and at this point, there may be more destabilisation incurred for the families as temporary accommodation could cease. The usual expectations are to secure work or study but often children of this group barely get food, clothes, or toys, and families rely heavily on charitable donations.

Telephone language services have been a long-term resource available in the UK. The language needs of the Afghan community are likely to be Pashto and Dari. In areas of high dispersal, the use of key workers from the same background would be useful in ensuring others in teams learn about the cultural and religious disposition of this community.

There may be a need to refer or to seek advice from safeguarding teams, where a child is in need or at risk. We cannot assume that the refugee status mitigates such referrals, and any impact on children is a usual way of life. Where there are needs not being met in the form of finances or a child is not able to get to school, for example, social

services have a duty of care to ensure both risk and "needs" are assessed under the Children's Act.

Trauma interventions in terms of what is an effect of resettlement, or witnessing events in Afghanistan recently or in the past, will require robust resources. Combat stress, a veteran support line, reported an exceedingly high number of calls from veterans who served in Afghanistan and required mental health support, including suicidal ideation due to post traumatic stress disorder. [3] Prioritising support within the NHS workforce, which has already been impacted by covid-19 will be quite a challenge. Additionally, services need to be aware that culturally many Afghan refugees will feel more at ease talking in a group setting, as opposed to individual support sessions. Grief is often seen as a community experience rather than an individual one. The need to develop community kinships and innovative methods to engage with these groups would be advisable.

Suffering, grief, and ambiguous loss will feature in many presentations, which will not be restricted to trauma. Separation from loved ones and death related losses will be part of the psychological needs of the refugees. Therefore, there may be a need to increase the service provision in primary care psychological services for trauma and complex grief, at local and national levels.

Health services and the health and social care workforce will need to be skilled and culturally ready to welcome Afghan refugees to the UK. The double burden of covid-19 and poor healthcare has impacted the already burdened healthcare system in Afghanistan. [4] After ensuring the population is registered with a local GP, another complicating factor in the present circumstances is the impact of covid-19. Health professionals will have to quickly establish the refugees' covid status, where they are in the vaccination programme, and introduce the refugees to covid-19 adjustments that the UK population has become used to. A big factor impacting the refugees will be their housing. If, as seems highly likely, they will be housed in overcrowded accommodation then this is the perfect breeding ground for covid-19 to spread. As stated, many will arrive in the UK in physically poor health, that in combination with substandard accommodation, and the threat of covid may prove fatal. Health professionals must be outspoken in trying to stop this scenario from unfolding.

The health and social care interface have the resources to support and ensure that a good value base is demonstrated through their practice. The disproportionate impact of covid-19 has already affected ethnic minorities across the globe. There is a small window of opportunity for services now to predict and assess their cultural responsiveness rather than rely solely on less appraised community-based initiatives.

Harjinder Kaur-Aujla, lecturer in mental health nursing, University of Birmingham.

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Psilocybin 6 Times More Effective Than Nicotine Patch to Help Smokers Quit

Healthline

New research suggests that psilocybin combined with CBT may be more effective than the nicotine patch for smoking cessation. Bloomberg Creative/Getty Images

Psilocybin, also known as “magic mushrooms,” may help smokers quit tobacco for the long term.

Smokers who received psilocybin alongside counseling were six times more likely to quit than those using nicotine patches in a small trial.

The findings suggest psilocybin could eventually become another tool to help millions of smokers quit.

Could “magic mushrooms” help tobacco smokers quit for good?

In a new trial, smokers treated with a combination of psilocybin and cognitive behavioral therapy (CBT) were significantly more likely to be abstinent 6 months after treatment than those using nicotine patches and CBT therapy.

The findings were published on March 10 in JAMA Network Open .

Over the past decade, research into the use of psychedelic drugs to treat psychiatric conditions has expanded dramatically. Psilocybin in particular has shown promise as a therapy for addiction-related conditions, including alcohol use disorder and smoking.

The new research suggests that psilocybin combined with CBT may be more effective than nicotine replacement therapy at helping smokers achieve both short- and long-term abstinence.

“There was no question the psilocybin group did much better,” said Matthew Johnson , PhD, professor of psychiatry and behavioral sciences and first author of the research.

“We found that for prolonged absence, they had six times higher odds of quitting if they were assigned to the psilocybin group compared to the nicotine patch group,” Johnson told Healthline.

Johnson is the leading researcher in the field of psilocybin and tobacco cessation. This latest trial builds on much of his earlier work, namely a 2014 study that first demonstrated that psilocybin could be safely used as an adjunct in smoking cessation treatment.

The findings published this week continue to establish psilocybin as a potential treatment for individuals with nicotine addiction.

Psilocybin users smoke-free at 6 months

Johnson and his team at Johns Hopkins University conducted a randomized clinical trial to investigate whether psilocybin could help smokers quit more effectively than a standard nicotine replacement therapy.

Both arms of the trial included 13 weeks of CBT, an approach used in counseling. CBT can help smokers identify triggers, manage cravings, and develop practical strategies to quit smoking.

The study enrolled 82 adult daily smokers ages 21 to 80 who had previously tried and failed to quit but still wanted to stop smoking . Participants were 59.8% male and 89% white, smoked roughly a pack per day, and had attempted quitting six times on average before the trial.

One group received a single supervised dose of psilocybin (30 milligrams per 70 kilograms of body weight), while the comparison group followed a standard 8- to 10-week regimen of FDA-approved nicotine patches.

Researchers tracked smoking behavior for six months using self-reported smoking diaries and biological tests to verify that participants had actually stopped smoking.

The researchers found that after starting treatment, 40% of those treated with psilocybin (17 participants) achieved prolonged abstinence, meaning they stopped smoking and remained smoke-free for six months after a brief grace period.

In contrast, only 10.0% of participants in the nicotine patch group (4 participants) achieved the same outcome. This translates to more than six times greater odds of quitting in the psilocybin group.

“It’s definitely refreshing to see someone look at new possibilities for nicotine addiction,” said George Singletary , MD, assistant professor of addiction medicine at the Tulane University School of Medicine, who wasn’t involved in the research.

“With all the deaths we have in this country due to nicotine that are preventable, it’s great to expand the options for treatment,” he told Healthline.

Can psilocybin lead to long-term smoking cessation?

The researchers also investigated a seven-day point-prevalence abstinence, meaning participants had not smoked at all in the previous week at the time of the six-month visit.

By this measure, 52% of the psilocybin group (22 participants) were abstinent compared with 25% of the nicotine patch group (10 participants), representing about three times higher odds of short-term abstinence.

“This is an exciting study,” said Brian Barnett , MD, assistant professor of psychiatry and director of the psychiatric treatment resistance program at the Cleveland Clinic, who wasn’t involved in the research.

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Doctors Thought Her Symptoms Were Pregnancy-Related. It Was Colon Cancer

Healthline

When 36-year-old Gabby Zappia (pictured above) reported blood in her stool, her doctor attributed it to pregnancy-related hemorrhoids. Months later, a colonoscopy revealed she had stage IV colon cancer. Gabby Zappia

Colorectal cancer in people under 50 is on the rise and is now the leading cause of cancer-related death for younger adults.

Experts say it's still unclear why cases are rising among people under 50.

Gabby Zappia is sharing her journey navigating diagnosis and treatment after her initial symptoms were misdiagnosed as pregnancy-related.

In 2024, Gabby Zappia was 36 years old and pregnant with her third child when she noticed blood in her stool.

"I brought it up to my OB, and she said it was likely pregnancy-related hemorrhoids. That explanation made sense, and I wanted it to make sense, so I trusted it," she told Healthline.

After her son was born, her symptoms persisted, and she pushed for answers.

"A colonoscopy changed my life overnight. Instead of finding hemorrhoids, they found a large mass in my colon," Zappia said.

In December 2024, Zappia was diagnosed with stage IV colon cancer that had spread to her liver.

"I was a full-time mom, managing all aspects of my kids' schedules, and I also had a small part-time job," she said. "After my diagnosis, I had to stop working to focus on appointments and recovery. My husband took over most of the day-to-day tasks that I had handled, and I had to step back significantly in my role as a mom."

Zappia immediately had a colon resection and, after recovering, started chemotherapy and immunotherapy in January 2025 at City of Hope.

In April 2025, she took a break from chemotherapy and underwent liver resection surgery and implantation of an HAI pump. Then she resumed chemotherapy after recovery.

"After 15 rounds of chemotherapy, I was declared no evidence of disease and rang the survivor bell in September 2025. A few months later, ctDNA tests showed cancer detection, and a PET scan confirmed activity in my liver," said Zappia.

She underwent another liver surgery in January 2026. Because her ctDNA remains detectable, she is now exploring clinical trials.

"Colon cancer is no longer just a disease of older adults, and it is on the rise. You know your body better than anyone. If something feels off, ask questions and request additional testing. Push for answers. Ask for the colonoscopy," Zappia said.

If you're not being heard, she stressed seeking a second opinion.

"We need more awareness. We need to listen to young patients. I am just one of many young faces of colon cancer, and if sharing my story helps even one person catch their cancer earlier, then sharing this journey has purpose," said Zappia.

Why is colorectal cancer in young adults on the rise?

Once considered an older person's disease, colorectal cancer is now the leading cause of cancer-related death in adults under 50.

According to a January 2026 JAMA study, colorectal cancer has surpassed breast and lung cancer to become the leading cause of cancer-related deaths in U. S. adults under 50.

Physicians at City of Hope, where Zappia received treatment, say they are now treating dozens of patients in their 20s, 30s, and 40s each week, reflecting what's happening nationwide.

Pashtoon Kasi, MD, MS, Medical Director of GI Medical Oncology at City of Hope Orange County, who treated Gabby, said three out of four people under the age of 50 are diagnosed with advanced disease.

"There are no screening guidelines for somebody below the age of 45. It's important to reiterate that the age of screening has moved from 50 to 45, [but] we're frequently seeing individuals in their 20s, 30s, 40s, and because there is no screening test when they're diagnosed, they're often advanced or metastatic," Kasi said.

While genetics can be a factor in a small percentage of early onset colorectal cancer, Kasi said the rise of colorectal cancer in younger people often occurs in people without any risk factors.

Researchers are looking into possible contributing factors, such as antibiotic use, the microbiome, diet, and microplastics, but no single factor explains the rise.

Paying attention to your body and symptoms is the strongest defense right now, said Kasi.

"A lot of our individuals, of course, they are young, so we've seen this cancer being diagnosed during or after pregnancy, and often it gets labeled as hemorrhoids or something that is not concerning, but in hindsight, probably should have warranted attention earlier," he said.

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Ultimate Guide to 4A Hair: Care Tips, Styles, and Maintenance

Greatist

Understanding your 4a hair type is essential for achieving healthy, vibrant curls. Discover the best care tips for 4a hair, from moisturizing routines to protective styles, and keep your curls looking their best.

Understanding your hair type is key to achieving healthy, luscious curls. Hair types range from 1 to 4 — straight, wavy, curly, and coily — with type 4 being the curliest. Type 4A hair has a fine texture and loose curl pattern, forming springy coils that need special care to stay healthy and vibrant.

Knowing your hair type helps determine the best hair care routine. Caring for 4A hair is essential, whether you're dealing with dryness or breakage or want to define your curls.

Let's dive into the world of 4A hair and discover how to keep those curls popping.

Catherine Falls Commercial/Getty Images

Alright, curl friends, let's break it down. Hair type falls into four main categories :

Type 1: Straight

Type 2: Wavy

Type 3: Curly

Type 4: Coily

Now, Type 4 hair is where things get seriously curly and coily . Within this category, we've got subtypes A, B, and C, which refer to the width and density of your curls.

So, what's the deal with 4A hair? 4A hair is the queen of the fine, loose curl pattern . It forms tight, springy coils that are like tiny, perfect S-shapes. These curls are less dense than their 4B and 4C cousins, making them more defined and delicate. Think of 4A hair as the Goldilocks of coily hair — not too tight, not too loose, but just right.

Your hair is unique and gorgeous, but it requires a bit of extra TLC to keep it looking its best . Here's a straightforward guide to keeping your 4A hair moisturized, healthy, and thriving:

Moisturize regularly

4A hair tends to be dry as its coily structure makes it difficult for natural oils to travel down the hair shaft. Keeping it moisturized is essential , so incorporate a good leave-in conditioner into your daily shampooing routine. Use deep conditioning treatments weekly to replenish moisture and strengthen your curls.

Don't forget scalp care , too! For maximum hydration, look for products with ingredients like shea butter, coconut oil, and glycerin.

Gentle detangling

4A hair is prone to tangles, so detangle gently to avoid breakage. Use a wide-tooth comb or your fingers to work through knots carefully. Start from the ends and work your way up to the roots, and always detangle when your hair is wet and coated with conditioner to provide slip and minimize damage.

Avoid heat

Heat styling tools can cause significant damage to 4A hair, leading to dryness, breakage, and loss of curl pattern. Minimize the use of blow dryers, flat irons, and curling wands. If you use heat, always apply a heat protectant spray first to shield your hair from high temperatures.

Pro tip: To maintain the health of your curls, opt for air-drying or styles that don't require heat.

Protective styles

Protective styles like braids, twists, and buns can help reduce manipulation and minimize breakage . These styles tuck away your ends, protecting them from friction and environmental damage.

Satin or silk pillowcases

Cotton pillowcases cause friction , which leads to frizz and breakage, especially for 4A hair. Switching to satin or silk pillowcases instead helps maintain

your curl pattern and prevent tangles.

Regular trims

Regular trims are crucial for keeping 4A hair healthy and preventing split ends from traveling up the hair shaft. Aim to trim your hair every 8-12 weeks to remove damaged ends and keep your curls looking fresh and vibrant.

Is 4A hair curly?

Yes, 4A hair is curly and falls under the coily category. It features tight, springy curls that form an S-shape.

Is my hair 4A, 4B, 4C, or 3C?

To determine your hair type, look at the curl pattern, texture, and density. 4A hair has a loose curl pattern with fine, springy coils, while 4B and 4C hair have tighter curls. 3C hair is curly but not as tight as Type 4.

How often should you wash 4A hair?

Washing once a week or every two weeks is generally sufficient for 4A hair. Overwashing can strip the hair of its

Pro tip: To maintain the health of your curls, opt for air-drying or styles that don't require heat.

Greatist

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Understanding Butt Cramps During Periods: Causes and Relief Tips

Greatist

Struggling with butt cramps during your period? You're not alone. Learn what causes this pain and how to find relief. From hormonal changes to anatomical factors, discover why your body acts up and what you can do.

Ah, the joys of "that time of the month" — cramping, bloating, and now, surprise butt cramps. As if menstrual pain wasn't enough, your body decides to throw in a curveball by extending the discomfort to your buttocks. But fear not! We've cracked the code on period butt cramps and how to deal with them.

Butt cramps during menstruation often feel like deep pelvic pains that shoot right down to your derrière. Butt why? It's all thanks to a few key players in your body's monthly routine .

Prostaglandins are hormone-like substances that prompt the uterine muscles to contract, helping the womb shed its lining. Although they're essential, they can go a bit overboard, causing the intense contractions to radiate downwards, causing that familiar, though unwelcome, ache in your buttocks.

The architecture of your anatomy also contributes. If your uterus tilts back toward your spine, it can push against nerves and amplify the pain in your booty when Aunt Flo comes to visit.

And let's not overlook endometriosis , a condition where tissue similar to the uterine lining grows outside the uterus. If it develops near the nerves that lead to the buttocks, it can exacerbate the situation, making the cramps even more severe and far-reaching.

While menstrual cycles are a frequent culprit behind butt cramps, the discomfort isn't always tied to your period . Several other factors can cause similar symptoms in peeps of all genders, including:

Muscle strain or injury: Engaging in strenuous physical activities or sports can lead to muscle strain in the pelvic and buttock areas.

Poor posture and prolonged sitting: Modern lifestyles can involve long periods of sitting, which stresses the pelvic region and buttocks.

Sciatica: Sciatic nerve pain can resemble period butt cramps.

Proctalgia fugax: This lesser-known but intense condition causes sudden cramping, spasming, or stabbing pain in the buttocks area.

Other conditions: Several other health issues can lead to a sensation of cramping in the booty, including muscle
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spasms, irritable bowel syndrome (IBS), intestinal issues , piles (hemorrhoids), and constipation .

Try these approaches to banish, or at least lessen, butt cramps during your period :

Heat therapy: Apply a heat pad or a hot water bottle to your lower back and buttocks for relief.

Exercise: While it might feel counterintuitive to move in pain , gentle physical activity can help.

Over-the-counter (OTC) pain relief: Non-prescription anti-inflammatory medications, such as ibuprofen, are often effective in managing menstrual cramps , including those that radiate to the buttocks.

Diet adjustments: Increasing your intake of magnesium-rich foods and omega-3 fatty acids may help.

If your butt cramps are severe, persistent, and impacting your quality of life, it's advisable to chat with a healthcare professional. Also, if cramps frequently occur outside of your menstrual cycle or are accompanied by other symptoms like abnormal bleeding or unusual discharge , call the doctor so they can rule out anything concerning.

If your period's cramping your style, you can take steps to ease booty and abdominal pain. Taming the beast below could be as simple as a hot water bottle, a brisk walk, or an omega-3 and magnesium-rich feast. So next time your body starts its monthly mutiny, try these tips and reclaim your comfort.

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React2Shell and related RSC vulnerabilities threat brief: early exploitation activity and threat actor techniques

Cloudforce One · Cloudflare Research

On December 3, 2025, immediately following the public disclosure of the critical, maximum-severity React2Shell vulnerability (CVE-2025-55182), the Cloudforce One Threat Intelligence team began monitoring for early signs of exploitation. Within hours, we observed scanning and active exploitation attempts, including traffic originating from infrastructure associated with Asian-nexus threat groups.

Early activity indicates that threat actors quickly integrated this vulnerability into their scanning and reconnaissance routines. We observed systematic probing of exposed systems, testing for the flaw at scale, and incorporating it into broader sweeps of Internet-facing assets. The identified behavior reveals the actors relied on a combination of tools, such as standard vulnerability scanners and publicly accessible Internet asset discovery platforms, to find potentially vulnerable React Server Components (RSC) deployments exposed to the Internet.

Patterns in observed threat activity also suggest that the actors focused on identifying specific application metadata — such as icon hashes, SSL certificate details, or geographic region identifiers — to refine their candidate target lists before attempting exploitation.

In addition to React2Shell, two additional vulnerabilities affecting specific RSC implementations were disclosed: CVE-2025-55183 and CVE-2025-55184. Both vulnerabilities, while distinct from React2Shell, also relate to RSC payload handling and Server Function semantics, and are described in more detail below.

Background: React2Shell vulnerability (CVE-2025-55182)

On December 3, 2025, the React Team disclosed a Remote Code Execution (RCE) vulnerability affecting servers using the React Server Components (RSC) Flight protocol. The vulnerability, CVE-2025-55182, received a CVSS score of 10.0 and has been informally referred to as React2Shell.

The underlying cause of the vulnerability is an unsafe deserialization flaw in the RSC Flight data-handling logic. When a server processes attacker-controlled payloads without proper validation, it becomes possible to influence server-side execution flow. In this case, crafted input allows an attacker to inject logic that the server interprets in a privileged context.

Exploitation is straightforward. A single, specially crafted HTTP request is sufficient; there is no authentication requirement, user interaction, or elevated permissions involved. Once successful, the attacker can execute arbitrary, privileged JavaScript on the affected server.

This combination of authenticated access, trivial exploitation, and full code execution is what places CVE-2025-55182

at the highest severity level and makes it significant for organizations relying on vulnerable versions of React Server Components.

In response, Cloudflare has deployed new rules across its network, with the default action set to Block. These new protections are included in both the Cloudflare Free Managed Ruleset (available to all Free customers) and the standard Cloudflare Managed Ruleset (available to all paying customers), as detailed below. More information about the different rulesets can be found in our documentation.

CVE

Cloudflare WAF Rule ID

CVE-2025-55182

React - RCE

Rules to mitigate React2Shell Exploit

Paid: 33aa8a8a948b48b28d40450c5fb92fba

Free: 2b5d06e34a814a889bee9a0699702280

CVE-2025-55182 - 2

React - RCE Bypass

Additional rules to mitigate exploit bypass

Paid: bc1aee59731c488ca8b5314615fce168

Free: cbdd3f48396e4b7389d6efd174746aff

CVE-2025-55182

Additional paid WAF rule to catch React2Shell scanning attempts

Paid: 1d54691cb822465183cb49e2f562cf5c

Recently disclosed RSC vulnerabilities

In addition to React2Shell, two additional vulnerabilities affecting specific RSC implementations were disclosed. The two vulnerabilities, while distinct from React2Shell, also relate to RSC payload handling and Server Function semantics, with corresponding Cloudflare protections noted below:

CVE

Cloudflare WAF Rule ID

CVE-2025-55183

In deployments where Server Function identifiers are insufficiently validated, an attacker may force the server into returning the source body of a referenced function, revealing the React Server Component ServerWorks name.

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Life lessons from the pain clinic

kellybrendel · The BMJ Blog

“Now is the start of the chronic phase.” This comment from a friend shortly after our baby son’s funeral grabbed my attention. As an anaesthetist, I appreciate the instant drug to effect gratification of anaesthesia for half of my working week. Yet pain medicine is my niche—specifically, the chronic, awkward pain that refuses to go away.

My clinic is filled with people who suffer in ways that are inconvenient to conventional medicine. They experience disparate pains that will not be boxed neatly inside a unifying diagnosis. Pain that comes with the kind of baggage it makes any attempts to cling to tidy Cartesian dualism seem delusional. I hope that my interactions with people facing such circumstances have been empathetic before now, although only they can judge. Faced with the chronic pain of grief for a child, I feel a new kinship with them.

Pain and grief are ugly things. We do not want them around because they make us afraid. We like to be in control, and loss and pain remind us that we never were. They blow apart our modern fantasy that successful living resides entirely in comfort and happiness. When this facade is shattered, difficult existential questions arise. These struggles and the questions they force upon us are most obvious among the dying: what does life mean in light of imminent death? What constitutes a “good” death, or a good life? How much will it hurt?

In chronic pain, the questions have a different flavour—one which more closely resembles the experience of those left behind, the bereaved ones. How do I continue to live this life I would never have chosen for myself? It really does hurt. Is it possible to make peace with a pain that has no obvious endpoint?

I have heard people in persistent pain express the desire to be able to put down their pain, or let someone else shoulder the burden, if only for a while. How I wish that were possible with grief. Both experiences are isolating. The desire to reach out to others for comfort is matched by overwhelming loneliness. At times, it feels like no one can reach me.

There are flare ups with grief, as with chronic pain. Some are predictable—a due date, an anniversary, an occasion that should be happy but now is not. These clouds gather visibly on the horizon—at least offering the opportunity to take shelter, to try to mitigate the blow. At other times they come out of a clear blue sky. We carry the weight around with us every day. On some days we are more equal to it than on others.

Hearing other people’s stories can be helpful. But as time passes, I am aware that there is a choice to make. It can be tempting to nurse grief, or pain, in an attempt to retain a connection with what has been lost. An identity can be found there, but it is one that is defined by loss. The journey from there to disabling bitterness is a short one. Instead,

can I choose to integrate the loss into the larger story of my life, and to build more layers of experience around it, which may insulate it a little in time?

Our culture does not deal well with pain. We are privileged to live in a society where diverse worldviews and faiths can be held freely. Yet we discuss them so rarely, when all is well in our private world, that in moments of shattering loss there is a risk that the fragile illusion of unity will break. We experienced, for example, the acute discomfort of those caring for our son when confronted with our faith. As a consequence, we have no social protocol for serious illness or death, and no shared paradigm. A Kurdish friend, on visiting us shortly after our son’s death, commented on the discomfort he felt on his first encounter with such a situation in British culture. We congratulated him on having become thoroughly enculturated—it’s always awkward, no one ever knows what to say. We simply appreciated the fact that he came.

At work I have often felt the discomfort of having little to add, no new wonder drug or procedure to guarantee a pain-free future for the person in front of me. Medical training conditions us to end each encounter with a plan—to categorise, then to manage. Over time I have learnt to sit on my hands, to acknowledge the discomfort but to keep my mouth shut and listen. I have long believed that this approach has value in its own right. As grieving parents we have encountered those who are too eager to “fix” us, albeit with the best of intentions and from a desire to make things better. In truth, there is no cure. I appreciate the people who continue to show up, and who are prepared to sit with us through the confusion and anger and sadness.

I truly admire the people I meet in clinic, who make the choice, most days at least, to accept their altered life and to find meaning in it in spite of the pain. These people keep on shouldering their particular burden, but refuse to be defined by it. I hope I can continue to learn from them.

Ruth Moore is an anaesthetist in the UK.

Competing interests: I have no financial interests, but in addition to my NHS appointment I am a volunteer doctor with Freedom From Torture.

The post Life lessons from the pain clinic appeared first on The BMJ .

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standard-article(title: [banana chocolate chip cake], author: [deb], source-name: [Smitten Kitchen], images: (), paragraphs: ([Somehow, despite how impossible it seems (to me, a person who has neither aged nor matured a day), it's been almost twenty years since I first told you about my family's favorite coffee cake . It's tall, plush, crisp with a flaky layer of cinnamon sugar on top, studded with a quilt of chocolate chips and is downright, well, adorable when cut into cubes because they're a little wobbly. When one tumbles, it shakes off a little pfft of cinnamon sugar, like a pup coming in from today's blizzard. It's perfect. It needs no changes or updates.], [Read more »],), insert-map: (:), word-count: 101, edited-for-length: false, debug-mode: false,)

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ANALYSIS

standard-article(title: [black bean confetti salad 2.0], author: [deb], source-name: [Smitten Kitchen], images: (), paragraphs: ([I was in Paris * last week — no, I cannot believe I get to utter sentences like that so casually, either, pinch me — and it was really, truly, and surprisingly spring. The magnolia trees at the Jardin du Palais Royal supplied us with a lace curtain of fluttering pink shadows, the daffodils and hyacinth were popping up from the ground like they'd missed us, and everyone was outside and stayed out until after midnight and this energy climbed inside me, evicted all of the seasonal malaise (turned out I was just cold!), and I did my best to bring all of this warmth and joy back to NYC with me. And despite the fact that my grouchy (sorry, “weathered”) friends tried to warn me that we were experiencing a “false spring” and “don't fall for it,” la la la, I said, it is spring in my heart now — and in my kitchen, and busted out a warm weather salad. Which is to say: I'm sorry, this sudden cold spell might be my fault.], [Read more »],), insert-map: (:), word-count: 180, edited-for-length: false, debug-mode: false,)

ScienceDaily Health, ScienceDaily Health: What teens eat might matter more for their mental health than previously thought. A sweeping review of nearly 20 studies found that healthier diets are often linked to fewer depressive symptoms, while poor eating habits may go hand in hand with greater psychological distress. Interestingly, focusing on whole dietary patterns—not just individual nutrients—showed more consistent benefits, suggesting that overall eating habits could play a meaningful role during this critical stage of brain development.

KC Morgan, Inhabitat: Among the oldest tools that human hands ever created are axes, used to chop wood for fires and buildings. Wood is truly the mother of sustainable resources, a material that will grow back again and again. And now, wood is becoming the “it” design material. Sweden is starting an ambitious new project that will demonstrate the majesty of wood, and the possibilities it represents in sustainable building.

ScienceDaily Health, ScienceDaily Health: Scientists have engineered probiotic bacteria to act as tumor-seeking drug factories. In mice, these bacteria infiltrated tumors and produced a cancer-fighting drug right where it was needed. This targeted approach could make treatments more effective and reduce side effects. More research is needed before it can be tested in people.

ScienceDaily Health, ScienceDaily Health: Antibiotics are accumulating in a major Brazilian river, especially during the dry season when pollution becomes more concentrated. Scientists even detected a banned drug inside fish sold for food, raising concerns about human exposure. A common aquatic plant showed promise in removing these chemicals from water—but it also altered how fish absorb them, creating unexpected risks.

ScienceDaily Health, ScienceDaily Health: Carrying extra fat around the waist may be more dangerous than the number on the scale suggests. Researchers found that belly fat was more strongly linked to heart failure risk than BMI, even in people with normal weight. Inflammation seems to play a key role, helping explain why this type of fat is especially harmful. Measuring waist size could offer a simple way to detect hidden risk earlier.

ScienceDaily, ScienceDaily: Scientists have discovered that the ocean’s “missing” plastic hasn’t vanished—it has broken down into trillions of invisible nanoplastics now spread through water, air, and living organisms. These tiny particles may be everywhere, including inside our bodies, raising serious concerns about their impact.

deb, Smitten Kitchen: class="smittenkitchen-print-hide">

Unless you're living your life better than me (probably!), I bet it's been way too long since you last had a baked potato for dinner — or, as they're more charmingly called across the pond, "jacket potato." And it's a crime because they're so cozy and uncomplicated to make, we could fix this right now.

[Read more »](#)

ScienceDaily, ScienceDaily: Scientists have created a new kind of carbon material that could make carbon capture much cheaper and more efficient. By carefully controlling how nitrogen atoms are arranged, they found certain structures capture CO₂ better and release it using far less heat. One version works at temperatures below 60 °C, meaning it could run on waste heat instead of costly energy. The discovery offers a powerful new blueprint for next-generation climate technology.

ScienceDaily Health, ScienceDaily Health: A new study suggests a widely used bone hormone could help relieve chronic back pain in an unexpected way. Instead of just strengthening bone, it appears to stop pain-sensing nerves from growing into damaged spinal areas. In animal models, this led to stronger spinal tissue and reduced pain sensitivity. The findings hint at a future treatment that tackles back pain at its biological roots.

Dawn Hammon, Inhabitat: If you love pasta, but don't often find suitable gluten-free alternatives in restaurants, Charlie's Table gluten-free products may be the answer for you. In fact, this craft pasta and pasta flour blend not only caters to the gluten intolerant, but is free of the top nine food allergens.

KC Morgan, Inhabitat: Nestled on the Rice University campus, amid multiple microhabitats and surrounded by green, you will see a small pavilion that looks a bit like an ancient Greek temple. But this pavilion was carefully crafted to have a modern design. It's sustainable, beautiful and an example that might just inspire campuses around the world.

ScienceDaily, ScienceDaily: Scientists have identified a key biological system that helps brown fat burn energy by building the networks it needs to function. A protein called SLIT3 splits into two parts, with each piece guiding the growth of blood vessels and nerves inside brown fat. These structures allow the tissue to pull in nutrients and rapidly convert them into heat instead of storing them as fat.

Scientific American, Scientific American: The past 11 years were the 11 hottest on record amid an increasing onslaught of climate-driven disasters, the World Meteorological Organization said in a new report

Scientific American, Scientific American: During their 10-day mission, this four-person crew will swing around the far side of the moon—and potentially travel farther from Earth than anyone in history

ScienceDaily Health, ScienceDaily Health: A major new U. S. cholesterol guideline is shifting the focus toward earlier, more personalized prevention of heart disease. It urges people to start screening sooner—sometimes even in childhood—and highlights the importance of tracking not just LDL (“bad”) cholesterol but also genetic risk factors like lipoprotein(a). A new, more advanced risk calculator now uses broader health data to better predict heart attack and stroke risk over decades.

Scientific American, Scientific American: A major defense against everything space throws at us, Earth's magnetic field may even protect the moon from damaging galactic cosmic rays

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The Wide Signal

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